CHE 230 HW2

1. a) $\sigma_e = \frac{\sigma}{A_0}$ $\qquad \varepsilon_e = \frac{L - L_0}{L_0}$ $\qquad \frac{L}{L_0} = 1 + \varepsilon_e$

$\sigma_T = \frac{F}{A}$ $\qquad V = A_0 L_0 = AL \rightarrow A = \frac{A_0 L_0}{L}$

$$\% RA = \left(\frac{A_0 - A}{A_0}\right)100 = \left(\frac{A_0 - \frac{A_0 L_0}{L}}{A_0}\right)100 = \left(1 - \frac{L_0}{L}\right)100$$

$$\% RA = \left(1 - \frac{1}{1+\varepsilon_e}\right)100$$

Greater % reduction in area for greater ε

Therefore $\boxed{\text{material B has greatest \% Reduction in Area}}$

b) D is strongest because it has the highest tensile strength. It can take the most stress before necking or fracturing

c) E is stiffest because it has the greatest elastic modulus. It elongates the least when stress is applied.

d) D is hardest because it has the highest tensile strength and highest fracture strength.

12

2. $E_{Tin} = 107\ GPa =$ $\qquad \sigma = E\frac{\Delta L}{L_0}$ $\qquad \Delta L = \frac{L_0 \sigma}{E}$

$\sigma = \frac{F}{A} = \frac{500\ N}{\pi(0.0015\ m)^2} = 7.07 \times 10^7 \frac{N}{m^2} = 0.0707\ GPa$

$L_0 = 2.5\ m$ $\qquad \Delta L = 1.65 \times 10^{-2}\ m$

$$\boxed{\Delta L = 16.5\ mm}$$

8

3 $\% CW = \frac{A_0 - A_d}{A_0} \times 100$

a $\% CW \; \frac{\pi\left(\frac{12^2}{4} - \frac{10.4^2}{4}\right)}{\pi \frac{12^2}{4}} \times 100 =$ 25 % CW

3

b	Yield Strength	Tensile Strength	Ductility
Copper	210 MPa	315 MPa	12 %
Brass	370 MPa	430 MPa	15 %
1040 steel	750 MPa	850 MPa	12 %

9

c. For high pressure, 1040 steel is most suited because it has the highest yield and tensile strength. It can withstand the greatest force without deforming or fracturing.

2

Brass is best for service where ductility is important. It can undergo the greatest plastic deformation before fracture.

4. $\tau_R = \sigma \cos\phi \cos\lambda = (61\ MPa)(\cos 56°)(\cos 32°) = 28.9\ MPa$

i) $\tau_R < \tau_{CRSS}$ so yielding will not occure

ii) $\sigma_y = \frac{\tau_{CRSS}}{\cos\phi\cos\lambda} = \frac{30.7\ MPa}{(\cos 56°)(\cos 32°)} = 64.7\ MPa$

64.7 MPa would be required for yielding to occur.

6

5. a. How are dislocations involved in the plastic deformation of metals/metal alloys? (4 points)

Plastic deformation occurs when sufficient stress is applied to a material. The bond an atom has with one of its neighbours is broken, and a new bond is formed through the dislocation. In effect, the dislocation moves along the slip plane one atom at a time. Dislocations allow deformation to occur without breaking all bonds simultaneously, which would require an extreme amount of energy.

b. Does the crystal structure of a metal affect its mechanical characteristics? If so, how and why (4 points)?

Yes. The crystal structure dictates how many slip systems are possible. FCC has 12 slip systems, so metals which have an FCC crystal structure are more ductile and are able to undergo plastic deformation more easily. HCP structures only have 3-6 slip systems, so metals with HCP crystal structure are more brittle. Grain size also affects mechanical strength, as different grain orientations make it more difficult for dislocations to slip across a grain boundary,

6. a. . How are mechanical properties affected by dislocation mobilities?(4 points)

Materials with high dislocation mobilities are able to experience slips more easily. As a result, they are able to undergo plastic deformation more easily and are more ductile.

4

b. What techniques are used to increase the strength/hardness of metals/alloys?(4 points)

Strain hardening (cold work) increases strength and hardness and reduces ductility in metals by reducing the cross sectional area at room temperature.

Solid solution hardening is the process of adding alloy elements that alter the lattice structure and make it more difficult for dislocations to move, making the metal stronger.

4

Grain size hardening reduces the size of grains and therefore increases the amount of grain boundaries. Grain boundaries are changes in the orientation of the crystal structure. More grain boundaries means there are more impediments to the motion of dislocations and the metal becomes stronger.

c. How are mechanical characteristics of deformed metal specimens altered by heat treatments? (4 points).

4

Heat treatment reverts metals to the properties before cold work. Hardness and strength are decreased and ductility is increased. During recovery, high temperature increases dislocation diffusion and new dislocations have low strain energy. During recrystallization, new grains are formed that consume the old grains, neutralizing their dislocations. The new grains have lower strain energy and more ductility. Finally, during grain growth, the size of grains increases. Grain boundaries store energy, so increasing the size of grains reduces the size of boundaries. Dislocations can move more freely when grain boundaries do not interrupt the structure, making the material less strong and more ductile. This entire process is called annealing.